

Presentation of project work

Two lessons in which the class does the teaching — and every mathematical claim is defended.

LESSON 51–52

2 × 40 MIN

GEOMETRY 10, LOYIHA ISHI

IGCSE E14.X

LESSON CLOCK

6'

50'

16'

12'

6'

The criteria	6 min
Presentations, five minutes each	50 min
Peer assessment	16 min
Whole-class review	12 min
The single lesson learned	6 min

LEARNING OBJECTIVES

- Present a piece of mathematical work clearly to an audience.
- Defend each measurement and each calculation under questioning.
- Assess another pair’s work against stated criteria.
- Identify one thing to do differently next time.

TEXTBOOK

National	pp. 239–242
Cambridge	Core & Extended, pp. 210–222
Workbook	Project sheet P3

01

Explanation

The criteria

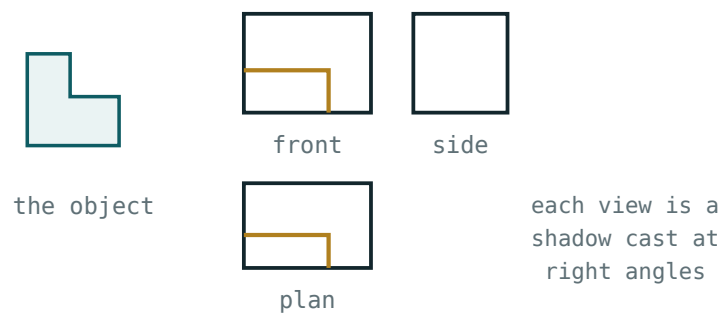
CRITERION	MARKS	WHAT EARNS THEM
the mathematics	6	correct theorems, correctly applied and named
the measurement or construction	4	done carefully, with the method described
the presentation	4	clear, audible, in five minutes, with a diagram
answering questions	4	defending a claim rather than repeating it
accuracy discussed	2	one honest sentence about the error

THE RULE FOR QUESTIONS

Every pair is asked at least two questions: one from the class and one from the teacher. At least one must be “*why is that angle a right angle?*” or “*how do you know that is the projection?*”

The three project types

TYPE	WHAT IS PRODUCED	THE MATHEMATICS TO DEFEND
scale model	a card building and its three orthogonal views	which lengths are preserved, and why
survey	a measured height from two angles of elevation	the plan triangle and the vertical triangle
demonstration	a physical model of the three perpendiculars	the statement, both directions



three orthogonal projections determine the solid

What a scale-model project has to produce, and get right.

Peer assessment, and the review

Each learner scores two other pairs on the five criteria and writes **one sentence** of specific praise and **one** specific suggestion. “Good work” scores nothing; “your plan view was aligned correctly under the elevation, which made it easy to read” scores full marks for the assessor.

THE WHOLE-CLASS REVIEW

Three questions, answered by the class together:

Which theorem was used most often across all the projects?

Which measurement was hardest to make accurately, and why?

Which presentation made its mathematics clearest, and what did it do?

Every learner then writes one line in the exercise book: “*Next time I will ...*”. Dated. It is read again at the end of Quarter IV.

02 Worked examples

EXAMPLE 1 A pair measured a tower as 47.5 m from 300 m and 9° . What should the accuracy sentence say?

- 1 The angle was measured to about $\pm 0.5^\circ$.
- 2 $300 \tan 8.5^\circ \approx 44.8$, $300 \tan 9.5^\circ \approx 50.2$.
- 3 So the height is between about 45 and 50 m.

ANSWER

“The height is 45–50 m; the angle dominates the error”

EXAMPLE 2 A pair claims their front elevation shows the true length of a sloping edge. Is that right?

- 1 Orthogonal projection preserves lengths **parallel** to the plane.
- 2 A sloping edge is not parallel to the front plane.
- 3 It appears foreshortened.

ANSWER

No — only lengths parallel to the plane of projection are preserved

EXAMPLE 3 A demonstration shows $c \perp OB$ and claims $c \perp AB$. What must be checked first?

- ① That AO really is perpendicular to the plane.
- ② That c lies in the plane and passes through B .
- ③ Only then does the theorem apply.

ANSWER

The two hypotheses of the theorem

03 Interactive model

Time each presentation strictly at five minutes; the discipline is part of the mark.

Assessing a project

Most marks are given for:

the model's appearance

the mathematics

the length of the talk

neatness

Orthogonal projection preserves lengths:

always

parallel to the plane

perpendicular to the plane

never

A survey needs:

one angle

two angles or one angle and a distance

three angles

no measurement

A useful peer comment is:

“good work”

specific about one thing

“I agree”

a mark only

An accuracy sentence should mention:

the answer

which measurement dominates the error

the weather

nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Project work	Loyiha ishi	Проектная работа
Presentation	Taqdimot	Презентация
Criteria	Mezonlar	Критерии
Peer assessment	O'zaro baholash	Взаимооценка
Justification	Asoslash	Обоснование
Measurement	O'lchash	Измерение
Scale model	Masshtabli model	Масштабная модель
Survey	O'lchov ishi	Съёмка
Conclusion	Xulosa	Заключение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

How many questions must each pair answer?

none

at least two

exactly one

ten

The accuracy criterion is worth:

0

2

6

10

Peer assessment requires:

a mark only

one praise and one suggestion

nothing written

a full essay

The lesson ends with:

a test

one dated sentence in the exercise book

homework

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. How long is each presentation?

ANSWER five minutes

2. How many criteria are there?

ANSWER 5

3. Total marks

ANSWER 20

4. Marks for the mathematics

ANSWER 6

5. Marks for accuracy discussed

ANSWER 2

6. Are questions part of the mark?

ANSWER yes, 4

7. What is written at the end?

ANSWER one dated "next time I will ..." sentence

Medium · the standard question

7 PROBLEMS

1. A tower from 300 m at $9^\circ \pm 0.5^\circ$: the range of heights

ANSWER ≈ 45 to 50 m

2. Does the front elevation show a sloping edge's true length?

ANSWER no

3. Which lengths does orthogonal projection preserve?

ANSWER those parallel to the plane

4. What must be checked before the three perpendiculars theorem?

ANSWER $AO \perp \alpha$, and $c \subset \alpha$ through B

5. A scale model at 1:50: a 12 m wall is how long?

ANSWER 24 cm

6. Same model: a 60 m² wall has what model area?

ANSWER 240 cm²

7. Which theorem is used most across the projects?

ANSWER the three perpendiculars

Hard · stretch and reason

7 PROBLEMS

1. A survey gives $h = d \tan \theta$. Which of d or θ usually dominates the error?

ANSWER θ — the tangent is sensitive near steep angles

2. A model at 1:50 uses 0.2 litres of paint. The real building needs?

ANSWER 500 litres

3. Show that a 0.5° error at 9° changes the height by about 6%

ANSWER $\frac{\Delta h}{h} \approx \frac{\Delta \theta}{\sin \theta \cos \theta}$

4. Why is a demonstration of the converse harder to build than the forward direction?

ANSWER It must start from a right angle you cannot see

5. Design a check that a measured height is right

ANSWER Measure from a second point and compare

6. A projection of a 60 m^2 roof at 35° : the plan area

ANSWER $\approx 49 \text{ m}^2$

7. Give one specific peer comment on a set of three views

ANSWER e.g. “your plan is aligned under the elevation, so the widths match”

07 Homework

Homework — 3 tasks

The dated sentence is read again at the end of Quarter IV.

1. Write your peer assessment of two projects: the five scores, one specific praise and one specific suggestion each.
2. Write your own “next time I will ...” sentence, dated.
3. Answer the three whole-class review questions in your own words.